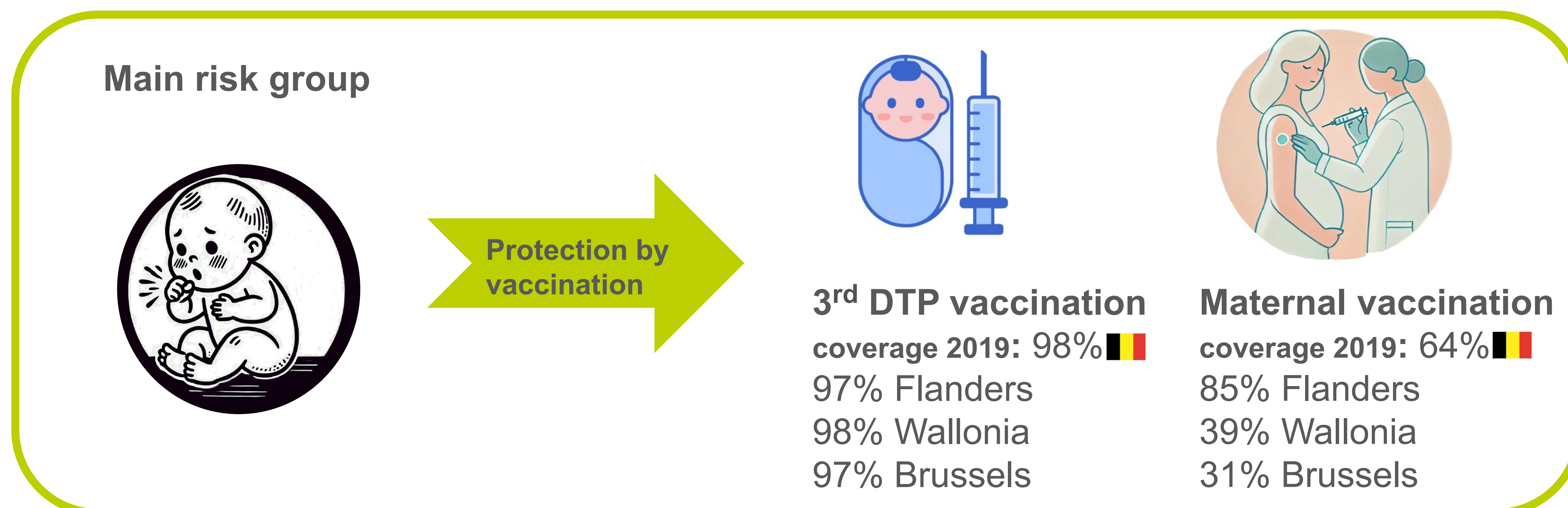


Disparities in maternal vaccination and infant pertussis in Belgium

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Introduction



We examine whether these regional disparities in maternal vaccination coverage are reflected in case numbers and severity among infant pertussis cases.

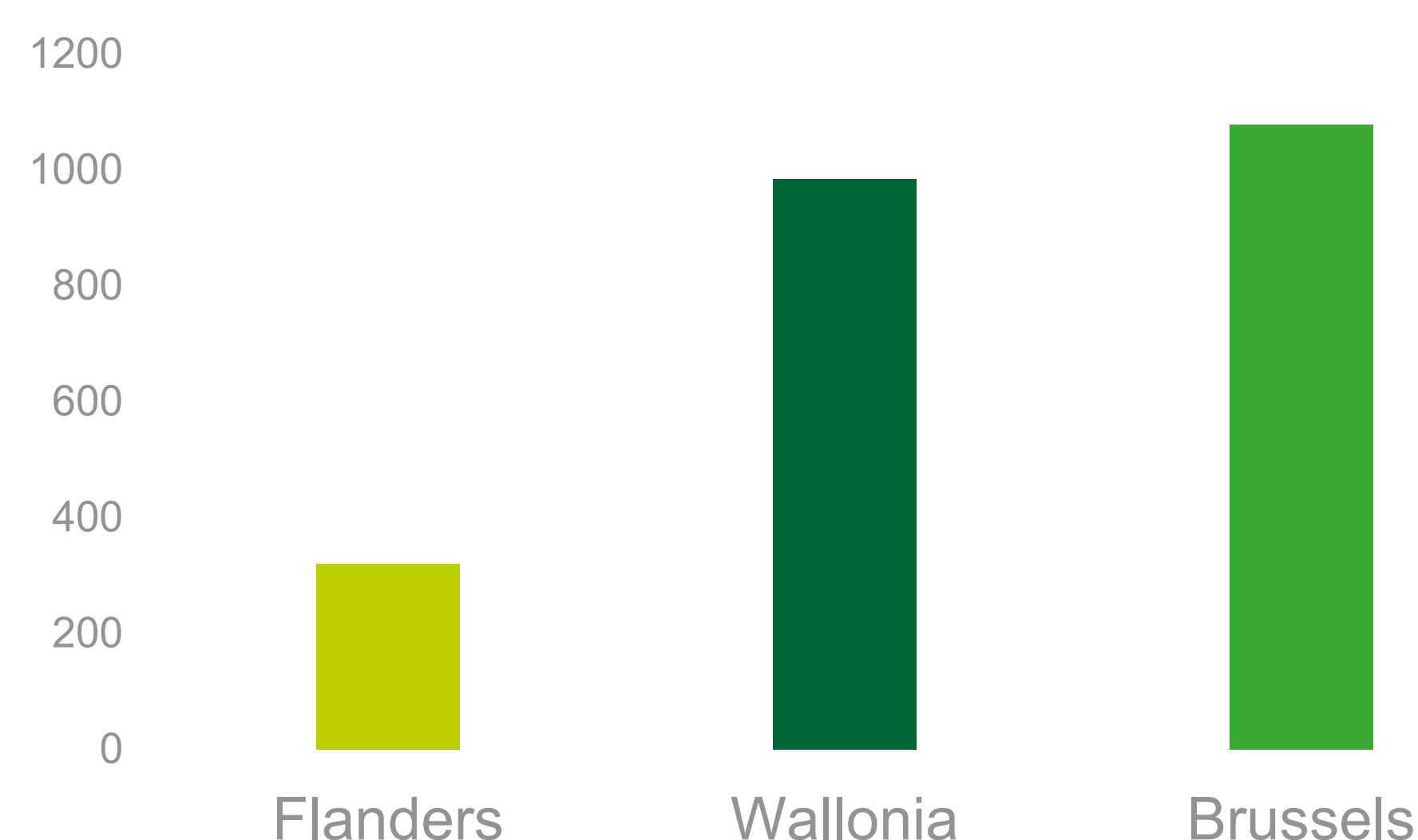
Methods

1. Number of pertussis cases reported to regional mandatory notifications (MN) in infants <8 weeks/ 100,000 and incidence rate ratios (IRR) for **2024**.
2. Regional hospitalization rates and ratios for pertussis in infants <1year for **2018-2020**, based on national administrative data.

Results

1. Incidence rate of pertussis cases <8 weeks by 100,000

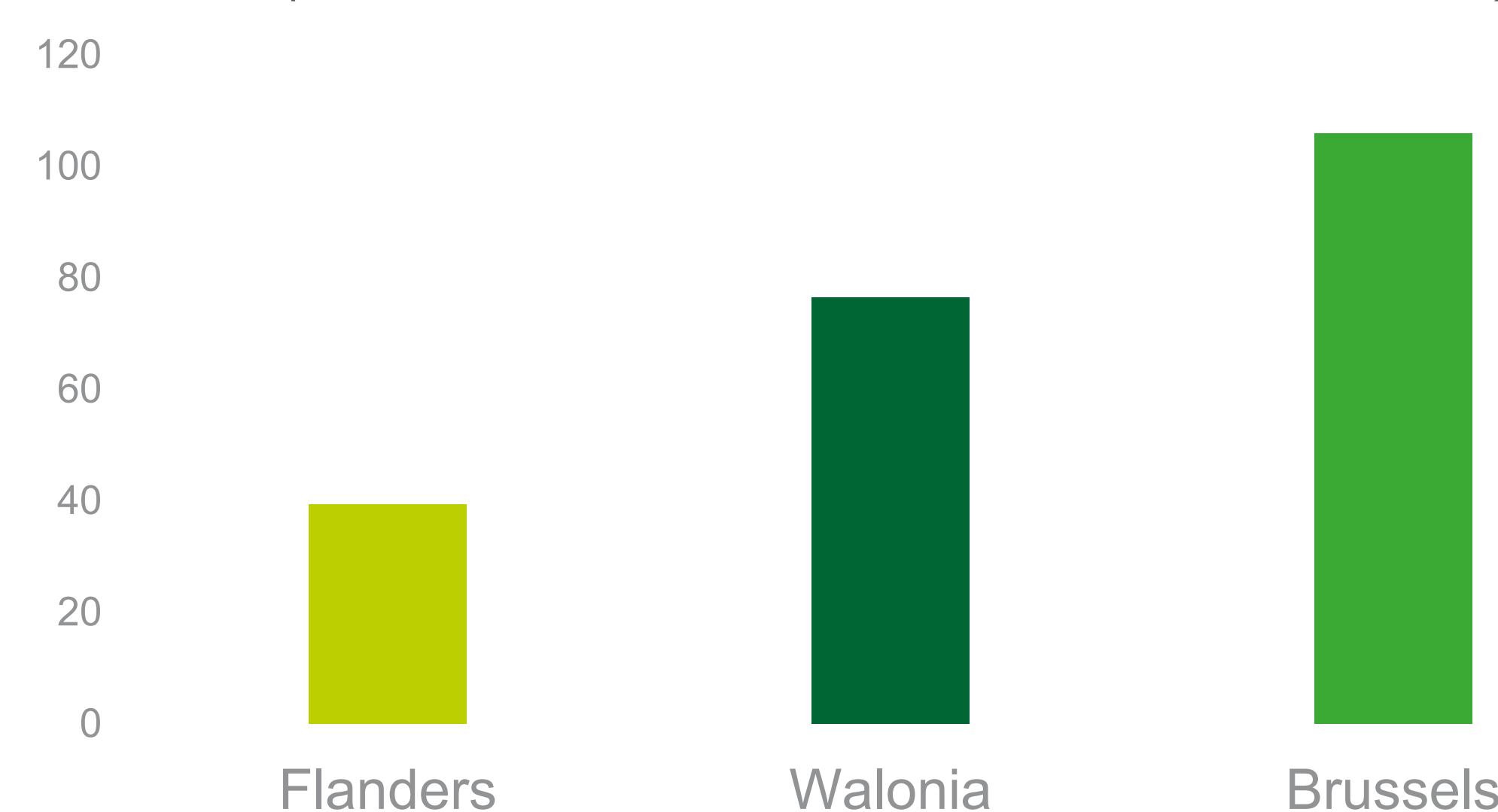
(source: MN, 2024)



	IRR (and 95% CI)
Bxl/FI	3.4 (2.0- 5.7)
Wal/FI	3.1 (2.0- 4.7)

2. Hospitalization rates for pertussis in infants <1year

(source: National administrative data, 2018-2020)



	IRR (and 95% CI)
Bxl/FI	2.7 (1.9- 3.8)
Wal/FI	1.9 (1.4- 4.2)

Independent data sources indicate higher pertussis burden among infants in Belgian regions with lower maternal vaccination coverage. Similar adult incidence and hospitalization rates across regions suggest true regional differences in infants. These findings highlight the need to address regional barriers to maternal vaccination.

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